**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

# LESSON PLAN

For

# AI for Network Security

TGS Ref No: TGS-2024051414

Conducted by

**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

**Version 1.2**


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## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes | Author |
|---|---|---|---|
| 1.0 | 29 August 2026 | Initial release of the AI for Network Security lesson plan — 2 days, 8 instructional hours per day, 4 topics aligned to LO1-LO4 and TSC ICT-DIT-4024-1.1, 12 hands-on AI-assisted labs, and the WA-SAQ + Case Study assessment on Day 2. | Dr. Alfred Ang |
| 1.1 | 29 August 2026 | Quality-audit revision. Process-map stage labels are now authored short names in the single source (never derived by truncation), with a build-time guard; procedure-slide headings use one uniform size per slide; the Support slide became a tile grid; the day schedule moved to a 9:30 am start with exactly 8 instructional hours per day (lunch and tea breaks excluded, enforced by assertion); the Case Study is 1 hour, matching the paper and the deck. | Dr. Alfred Ang |
| 1.2 | 29 August 2026 | Wired the browser practice tools into the lab procedures. The IP Calculator now scopes the scan in Lab 1 and carves the security zones in Lab 7; the PCAP Analyzer triages the capture in Lab 2 and summarises capture evidence in Lab 11; the Cybersecurity Simulator grounds the likelihood scores in Lab 5; the Cryptography Toolkit demonstrates the confidentiality control in Lab 8 and hash | Dr. Alfred Ang |


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| | | tamper-evidence in Lab 11; RegexLab builds and negatively tests the log filters in Lab 10. Hacklab and FauxBank are documented as optional, unassessed extension work. | |


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## Course Information

| | |
|---|---|
| **Course Title** | AI for Network Security |
| **WSQ Course Reference** | TGS-2024051414 |
| **Training Provider** | Tertiary Infotech Academy Pte Ltd (UEN 201200696W) |
| **Duration** | 2 days · 8 instructional hours per day (16 hours) |
| **Daily Timing** | Day 1: 9:30 am – 7:00 pm · Day 2: 9:30 am – 7:00 pm (1-hour lunch and two 15-minute tea breaks, excluded from the 8 instructional hours) |
| **Mode** | Instructor-led, hands-on AI-assisted network security labs against authorised lab targets |
| **Skills Framework** | TSC: Network Security (ICT-DIT-4024-1.1) |
| **Trainer** | Dr. Alfred Ang |

## Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Identify and assess network threats and vulnerabilities by analysing ICT configurations.
- LO2: Evaluate organisational assets and develop risk management plans to address network infrastructure weaknesses effectively.
- LO3: Design and implement security measures for network components by managing security phases.
- LO4: Document security incidents and apply incident response configurations to manage risks based on established procedures.

## Assessment

- Written Assessment — Short-Answer Questions (WA-SAQ) · 1 hour · open book. Assesses the underpinning knowledge K1–K4.
- Case Study (CS) · 1 hour · open book. Assesses the abilities A1–A9 against a realistic organisational scenario.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 5:00 pm.
- Knowledge (K1-K4) is assessed by the WA-SAQ; abilities (A1-A9) are assessed by the Case Study.
- A minimum of 75% attendance is required to be eligible for assessment and funding.


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## Course Schedule

### Day 1 — AI-Assisted Threat & Vulnerability Analysis · AI-Powered Risk Assessment and Management Planning

| Time | Duration | Topic / Activity | Slides |
|---|---|---|---|
| 9:30–10:00 | 30 min | Welcome, course introduction, ground rules, learning outcomes and mandatory digital attendance (AM) | 1-19 |
| **10:00–11:15** | 75 min | **Core concepts — network security and the CIA triad; why signature-based detection fails against zero-day, polymorphic and Living-off-the-Land attacks; where AI is applied in network security (concepts + demo)** | 20-31 |
| 11:15–11:30 | 15 min | Tea break | |
| 11:30–13:30 | 120 min | Topic 1 — AI-Assisted Network Threat and Vulnerability Analysis: ICT networks and configuration (K1), the threat landscape (K2), pattern-of-life baselines, NDR and AI vulnerability prioritisation (concepts). Hands-on: Lab 1: Map the ICT Network and Analyse Its Configuration; Lab 2: Baseline Normal Traffic and Detect an Anomaly with AI | 32-51 |
| 13:30–14:30 | 60 min | Lunch break | |
| 14:30–16:15 | 105 min | Hands-on: Lab 3: AI-Assisted Vulnerability Analysis and Prioritisation. Topic 2 — AI-Powered Network Risk Assessment and Management Planning: asset discovery and classification, CIA business impact (concepts) | 52-66 |
| 16:15–16:30 | 15 min | Tea break | |


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| 16:30–19:00 | 150 min | Topic 2 continued — risk scoring on the 5x5 matrix, the four treatments, NIST CSF 2.0 and the NIST AI RMF (concepts). Hands-on: Lab 4: Build an Asset Register and Classify What Needs Protecting; Lab 5: AI-Assisted Risk Scoring and the Risk Register; Lab 6: Develop the Risk Management Plan. Day 1 recap, Q&A and PM digital attendance | 67-83 |

*Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch and the two 15-minute tea breaks.*

## Day 2 — Designing & Implementing AI-Enhanced Security Measures · AI-Assisted Incident Documentation and Response

| Time | Duration | Topic / Activity | Slides |
|---|---|---|---|
| 9:30–9:45 | 15 min | Day 1 recap and mandatory digital attendance (AM) | |
| **9:45–11:15** | 90 min | **Topic 3 — Designing and Implementing AI-Enhanced Network Security Measures: defence in depth and the three control categories; the planning, building and management phases; segmentation, zero trust and access control (K3, concepts + demo)** | 84-96 |
| 11:15–11:30 | 15 min | Tea break | |
| 11:30–13:30 | 120 min | Topic 3 continued — AI-enhanced firewalls, IDS/IPS, UEBA and adaptive access control; securing the AI itself (NIST AI 100-2 attack taxonomy, NSA/CISA deployment guidance, MITRE ATLAS). Hands-on: Lab 7: Design the Layered Architecture Across the Three Security Phases; Lab 8: Implement AI-Enhanced Controls — | 97-115 |


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| | | Firewall Policy, IDS and Adaptive Access; Lab 9: Secure the AI Itself — Adversarial ML and the NSA/CISA Deployment Controls | |
| 13:30–14:30 | 60 min | Lunch break | |
| 14:30–16:00 | 90 min | Topic 4 — AI-Assisted Security Incident Documentation and Response: logging and audit design, SIEM/SOAR, AI alert triage, the NIST SP 800-61 lifecycle and incident documentation standards (K4, concepts). Hands-on: Lab 10: Centralise Logs and Triage Alerts with AI; Lab 11: Document a Security Incident to the Standard | 116-138 |
| 16:00–16:15 | 15 min | Tea break | |
| 16:15–16:45 | 30 min | Hands-on: Lab 12: Implement the Incident Response Configuration and Playbook. Course recap, TRAQOM survey and assessment digital attendance | 139, 146-151 |
| **16:45–17:00** | 15 min | **Briefing for Assessment** | 13 |
| **17:00–18:00** | 60 min | **Written Assessment — Short-Answer Questions (WA-SAQ), 1 hour, open book. Assesses the underpinning knowledge K1-K4** | 152-153 |
| **18:00–19:00** | 60 min | **Case Study (CS), 1 hour, open book. Assesses the abilities A1-A9 against a realistic organisational scenario** | 152-154 |

*Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch and the two 15-minute tea breaks.*


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## Lab Reference (aligned to learning outcomes and TSC statements)

| Topic / Learning unit | Weighting | K / A assessed | Labs | Slides |
|---|---|---|---|---|
| **Topic 01: AI-Assisted Network Threat and Vulnerability Analysis** | 25% | K1 \| A1, A2 | Lab 1, Lab 2, Lab 3 | Slides 41-56 |
| **Topic 02: AI-Powered Network Risk Assessment and Management Planning** | 25% | K2 \| A3, A4 | Lab 4, Lab 5, Lab 6 | Slides 67-82 |
| **Topic 03: Designing and Implementing AI-Enhanced Network Security Measures** | 30% | K3 \| A5, A6, A7 | Lab 7, Lab 8, Lab 9 | Slides 97-114 |
| **Topic 04: AI-Assisted Security Incident Documentation and Response** | 20% | K4 \| A8, A9 | Lab 10, Lab 11, Lab 12 | Slides 127-144 |


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